JANMITRA

Civic Issue Reporting & Management System

Nagpur Municipal Corporation | 4 Categories | 4 Wards | 4 Departments

Team ResolveIT

Worker Management & Ward Assignment

Worker registration, ward assignment, and complaint routing rules

CITIZEN PORTAL	Report Issue | Track Status | Verify Resolution
BACKEND (Django)	REST API | Auth | Business Logic | AI Module
AI VALIDATION	Fake Detection (EXIF+ELA+HuggingFace) | YOLO Relevance Check
DATABASE	PostgreSQL + PostGIS | 4 Categories | 4 Wards | 4 Depts
DEPT. DASHBOARD	PWD | Solid Waste | Water Works | Electrical

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1. Worker System Overview

Every NMC field worker in JANMITRA is linked to exactly ONE department and exactly ONE ward. This two-key assignment ensures that complaints automatically reach the right worker based on what type of issue it is (category -> department) and where it is (ward). The system prevents mismatched assignments by default, while allowing admin overrides when genuinely needed.

```
WORKER ASSIGNMENT LOGIC:

complaint.category  ->  complaint.department_id  (auto, from category master)
complaint.ward      ->  complaint.ward_id         (citizen selected)

Available workers for this complaint =
    User.objects.filter(
        role        = 'worker',
        department  = complaint.department,   <-- MUST MATCH
        ward        = complaint.ward,          <-- MUST MATCH
        status      = 'active',
    )

Exception: Admin can set ward_override=True with a reason text.
           This is logged in status_history for audit.
```

2. Worker Registration Fields

Field	Type	Required	Notes
Worker ID	Auto UUID	Auto	Generated by system on registration
Employee ID	VARCHAR(50)	Yes	NMC employee number (must be unique)
Full Name	VARCHAR(100)	Yes	Worker's full name
Mobile Number	VARCHAR(15)	Yes	Used as login username (must be unique)
Email	VARCHAR(255)	No	Optional contact email
Department	FK -> departments	Yes	One of 4: PWD / SWM / PHE / Electrical
Ward	FK -> wards	Yes	One of 4: Ward 1 / Ward 2 / Ward 3 / Ward 4
Designation	VARCHAR(100)	Yes	E.g.: Field Inspector, Supervisor, Technician
Status	VARCHAR(20)	Auto	active (default) / inactive / on_leave
Password	TEXT (hashed)	Yes	Initial password set by admin (worker changes later)
FCM Token	TEXT	Auto	Set by app on first login (push notifications)
Registered At	TIMESTAMP	Auto	System-generated registration timestamp

3. Ward + Department Combinations

JANMITRA has 4 departments and 4 wards. That means 16 possible department+ward combinations. Each combination should ideally have at least 1 active worker registered.

Dept. \ Ward	Ward 1	Ward 2	Ward 3	Ward 4
PWD	Min 1 worker	Min 1 worker	Min 1 worker	Min 1 worker
SWM	Min 1 worker	Min 1 worker	Min 1 worker	Min 1 worker
PHE/Water	Min 1 worker	Min 1 worker	Min 1 worker	Min 1 worker
Electrical	Min 1 worker	Min 1 worker	Min 1 worker	Min 1 worker

TIP Seed Data: The demo seed command creates at least 1 worker per dept+ward combination = 16 demo workers minimum. This ensures the system can demonstrate assignment for any complaint.

4. Worker Registration Process

4.1 Who Can Register Workers

Admin	Can register workers for any department and any ward. Full access.
Dept. Officer	Can register workers only for their own department. Can select any ward for those workers.
Worker	Cannot register other workers. Self-registration not allowed -- admin or officer creates account.
Citizen	Cannot register as a worker. Separate account type.

4.2 Registration API

```
POST /api/v1/admin/workers/
Permission: IsAdmin | IsDeptOfficer

Request body:
{
  'employee_id': 'NMC2024001',
  'full_name':   'Suresh Patil',
  'mobile':      '9876543210',
  'email':       'suresh@nmc.gov.in',  // optional
  'department':  2,                    // SWM dept ID
  'ward':        1,                    // Ward 1
  'designation': 'Field Inspector',
  'password':    'initial_password_123'
}

Validation:
  - mobile must be unique across all users
  - employee_id must be unique
  - DeptOfficer: department must equal request.user.department_id
  - ward must be 1, 2, 3, or 4

Response 201: {id, employee_id, full_name, mobile, department, ward, designation}
```

4.3 Worker Profile Update

```
PATCH /api/v1/admin/workers/{id}/
Permission: IsAdmin | IsDeptOfficer (own dept only)

Allowed updates: designation, status (active/inactive/on_leave), email, ward
Note: Changing a worker's ward or department is an admin-only action.
      System logs the change with timestamp and changed_by user.

Restricted: mobile and employee_id cannot be changed once set.
            If error, admin must deactivate old account and create new one.
```

5. Complaint Assignment Rules

5.1 Standard Assignment (Dept + Ward Match)

Who assigns	Dept. Officer (for their dept) or Admin
Available workers	API returns: workers where dept == complaint.dept AND ward == complaint.ward AND status == active
Officer selects	Chooses from available list. Clicks 'Assign'.
System validates	Backend re-validates the match before saving (prevents race conditions)
On success	complaint.assigned_worker_id set. Status -> Assigned. Worker gets push notification.

5.2 Admin Override Assignment

When used	No workers available in required dept+ward, or emergency cross-ward reassignment
Who can do it	Admin only
UI	Admin sees all workers in dropdown. Ward mismatch shown as yellow warning badge.
Required	Override checkbox must be checked. Reason text field is mandatory (min 10 chars).
System records	complaint.ward_override = True, complaint.override_reason = reason text, status_history entry created with admin's name and reason
Audit trail	Override visible to all officers viewing the complaint detail. Cannot be hidden.

5.3 Database Constraint (Safety Net)

```
-- Partial constraint: only enforce when ward_override is False
-- This is implemented at the application layer (validators.py)
-- and documented here for clarity:

-- When ward_override = FALSE:
--   worker.ward_id  MUST EQUAL complaint.ward_id
--   worker.dept_id  MUST EQUAL complaint.dept_id

-- When ward_override = TRUE:
--   No constraint on ward or dept match
--   But override_reason MUST be non-empty

-- Application-level check in complaints/validators.py:
-- validate_worker_assignment(worker, complaint, override=False)
```

6. Worker Task View (What Worker Sees)

When a worker logs in, they see only complaints from their department + ward:

```
GET /api/v1/worker/tasks/
-- Always filtered by:
--   complaint.department_id = request.user.department_id
--   complaint.ward_id       = request.user.ward_id
--   complaint.assigned_worker_id = request.user.id  (or unassigned)
--   status NOT IN [Closed, Invalid, Rejected -- Fake Image]
-- Ordered by: priority_score DESC (Critical first)

Returns: [{id, category, ward, description, photos, location,
           priority_score, priority_level, status, submitted_at}]
```

7. Worker Management Dashboard (Admin View)

Admin Worker Management page shows all workers across all departments and wards:

Column	Content	Filterable?
Employee ID	NMC employee number	No
Name	Worker full name	Yes (search)
Mobile	Login mobile number	No
Department	PWD / SWM / PHE / Electrical	Yes (dropdown)
Ward	Ward 1 / 2 / 3 / 4	Yes (dropdown)
Designation	Field Inspector, Supervisor, etc.	No
Active Tasks	Number of In-Progress complaints	Yes (sort)
Status	Active / Inactive / On Leave	Yes (dropdown)
Registered	Registration date	Yes (sort)
Actions	Edit | Deactivate	--